

Assignment No 10

AIM: Develop a Java application that connects to a MySQL/PostgreSQL database to manage student information. Implement CRUD operations for student records using JDBC. Use prepared statements to handle SQL queries securely and ensure proper transaction management.

Code:

```
import java.sql.*;
import java.util.Scanner;

public class StudentCURD {

    public static void main(String[] args) {

        try {
            Conn conn = new Conn();
            Scanner sc = new Scanner(System.in);

            int choice;

            do {
                System.out.println("\n===== STUDENT MENU =====");
                System.out.println("1. Display All Students");
                System.out.println("2. Insert Student");
                System.out.println("3. Update Student Address");
                System.out.println("4. Delete Student");
                System.out.println("5. Search Students (ID Range)");
                System.out.println("6. Exit");
                System.out.print("Enter choice: ");
                choice = sc.nextInt();

                switch (choice) {

                    case 1:
                        ResultSet rs = conn.s.executeQuery("SELECT * FROM student");

                        System.out.println("\nID Name Age Email Address");

                        while (rs.next()) {
                            System.out.println(
                                rs.getInt("id") + " " +
                                rs.getString("name") + " " +
                                rs.getInt("age") + " " +
                                rs.getString("email") + " " +
                                rs.getString("address")
                            );
                        }
                    }
                }
            }
        }
    }
}
```

```
}  
break;
```

case 2:

```
System.out.print("Enter ID: ");  
int id = sc.nextInt();  
sc.nextLine();
```

```
System.out.print("Enter Name: ");  
String name = sc.nextLine();
```

```
System.out.print("Enter Age: ");  
int age = sc.nextInt();
```

```
System.out.print("Enter Email: ");  
String email = sc.nextLine();
```

```
System.out.print("Enter Address: ");  
String address = sc.nextLine();
```

```
String insert = "INSERT INTO student VALUES (" + id + ", " + name + ", "  
age + ", " + email + ", " + address + ")";  
conn.s.executeUpdate(insert);
```

```
System.out.println("Student Inserted!");  
break;
```

case 3:

```
System.out.print("Enter ID to update: ");  
int uid = sc.nextInt();  
sc.nextLine();
```

```
System.out.print("Enter New Address: ");  
String newAddress = sc.nextLine();
```

```
String update = "UPDATE student SET address='" + newAddress + "' WHERE  
id=" + uid;  
conn.s.executeUpdate(update);
```

```
System.out.println("Address Updated!");  
break;
```

case 4:

```
System.out.print("Enter ID to delete: ");  
int did = sc.nextInt();
```

```
String delete = "DELETE FROM student WHERE id=" + did;
conn.s.executeUpdate(delete);
```

```
System.out.println("Student Deleted!");
break;
```

```
case 5:
```

```
System.out.print("Enter Start ID: ");
int start = sc.nextInt();
```

```
System.out.print("Enter End ID: ");
int end = sc.nextInt();
```

```
ResultSet rs1 = conn.s.executeQuery(
    "SELECT * FROM student WHERE id BETWEEN " + start + " AND " +
end
);
```

```
System.out.println("\nID Name Age Email Address");
```

```
while (rs1.next()) {
    System.out.println(
        rs1.getInt("id") + " " +
        rs1.getString("name") + " " +
        rs1.getInt("age") + " " +
        rs1.getString("email") + " " +
        rs1.getString("address")
    );
}
break;
```

```
case 6:
```

```
System.out.println("Exiting...");
break;
```

```
default:
```

```
System.out.println("Invalid choice!");
```

```
}
} while (choice != 6);
```

```
conn.c.close();
```

```
} catch (Exception e) {
    e.printStackTrace();
}
}
}
```

Output:

```
===== STUDENT MENU =====
```

1. Display All Students
2. Insert Student
3. Update Student Address
4. Delete Student
5. Search Students (ID Range)
6. Exit

Enter choice: 1

ID	Name	Age	Email	Address
1	Amit Patil	18	amit@gmail.com	Pune
2	Rohit Deshmukh	19	rohit@gmail.com	Mumbai
3	Sneha Kulkarni	18	sneha@gmail.com	Kolhapur
4	Priya Joshi	20	priya@gmail.com	Nashik
5	Kunal Jadhav	19	kunal@gmail.com	Satara
6	tejas shinde	18	neha@gmail.com	Solapur
7	Akash More	20	akash@gmail.com	Pune
8	Pooja Pawar	19	pooja@gmail.com	Sangli
9	Vishal Patil	18	vishal@gmail.com	Kolhapur
10	Anjali Kadam	20	anjali@gmail.com	Mumbai
11	Omkar Chavan	19	omkar@gmail.com	Pune
12	Riya Desai	18	riya@gmail.com	Surat
13	Saurabh Kulkarni	20	saurabh@gmail.com	Nagpur
14	Mayuri Patil	19	mayuri@gmail.com	Jalgaon
15	Aditya Joshi	18	aditya@gmail.com	Aurangabad
16	Shruti More	20	shruti@gmail.com	Pune

```
===== STUDENT MENU =====
```

1. Display All Students
2. Insert Student
3. Update Student Address
4. Delete Student
5. Search Students (ID Range)
6. Exit

Enter choice: 2

Enter ID: 102

Enter Name: Om Hire

Enter Age: 18

Enter Email: Enter Address: lohegoan pune

Student Inserted!

```
===== STUDENT MENU =====  
1. Display All Students  
2. Insert Student  
3. Update Student Address  
4. Delete Student  
5. Search Students (ID Range)  
6. Exit  
Enter choice: 3  
Enter ID to update: 102  
Enter New Address: Katraj Pune  
Address Updated!
```

```
===== STUDENT MENU =====  
1. Display All Students  
2. Insert Student  
3. Update Student Address  
4. Delete Student  
5. Search Students (ID Range)  
6. Exit  
Enter choice: 4  
Enter ID to delete: 102  
Student Deleted!
```

===== STUDENT MENU =====

1. Display All Students
2. Insert Student
3. Update Student Address
4. Delete Student
5. Search Students (ID Range)
6. Exit

Enter choice: 5

Enter Start ID: 10

Enter End ID: 20

ID	Name	Age	Email	Address
10	Anjali Kadam	20	anjali@gmail.com	Mumbai
11	Omkar Chavan	19	omkar@gmail.com	Pune
12	Riya Desai	18	riya@gmail.com	Surat
13	Saurabh Kulkarni	20	saurabh@gmail.com	Nagpur
14	Mayuri Patil	19	mayuri@gmail.com	Jalgaon
15	Aditya Joshi	18	aditya@gmail.com	Aurangabad
16	Shruti More	20	shruti@gmail.com	Pune
17	Nikhil Pawar	19	nikhil@gmail.com	Sangli
18	Komal Jadhav	18	komal@gmail.com	Satara
19	Tejas Shinde	20	tejas@gmail.com	Beed
20	Rahul Patil	19	rahul@gmail.com	Kolhapur

===== STUDENT MENU =====

1. Display All Students
2. Insert Student
3. Update Student Address
4. Delete Student
5. Search Students (ID Range)
6. Exit

Enter choice: 6

Exiting...